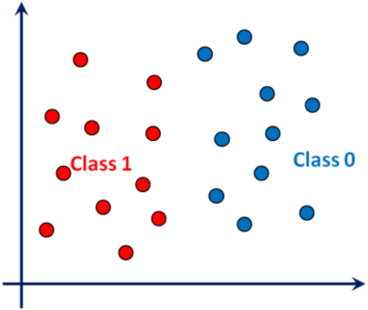
Support Vector Machines

Supervised Learning Model

Linear Classifier

- Decision boundary: Hyperplane
- Class 1 lies on the positive side
- Class 0 lies on the negative side



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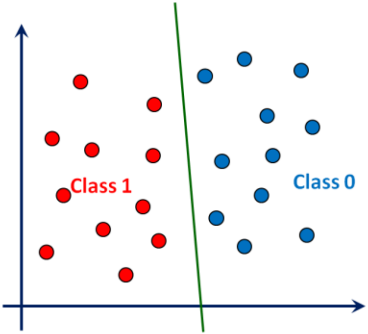


Linear Classifier

- Decision boundary: Hyperplane

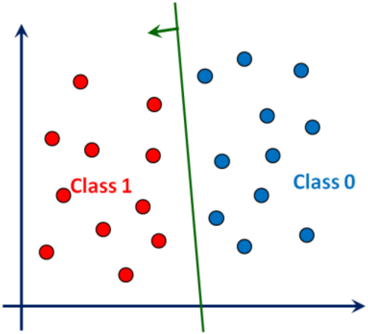
$$w^T x = 0$$

- Class 1 lies on the positive side

$$w^T x > 0$$

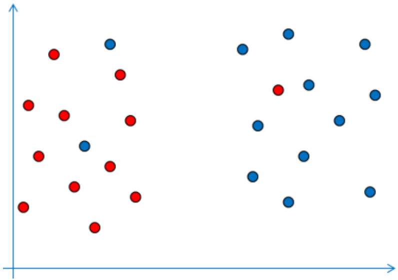
- Class 0 lies on the negative side

$$w^T x < 0$$



Why Linear? Generalization vs. Complexity

- Is it good to use a complex curve to reduce training error?



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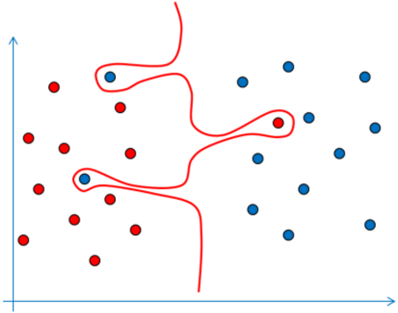

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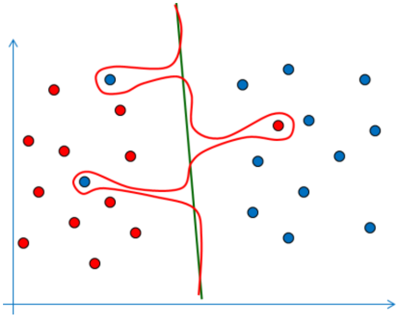

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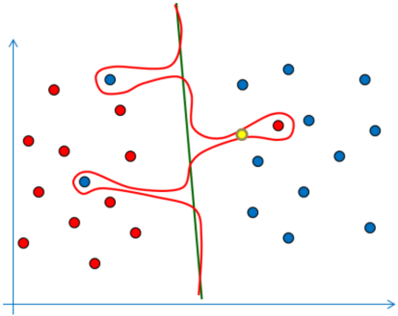
Are both solutions equally good?



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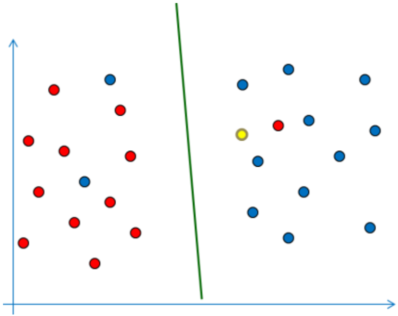

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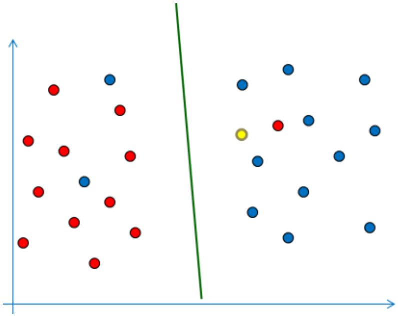
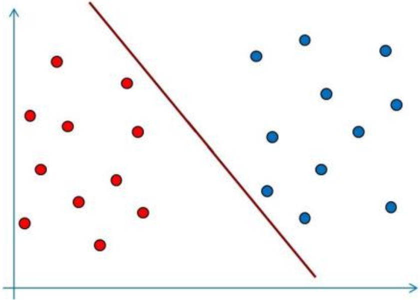


Summary

- Linear Classifiers are simple and hence efficient
- There exists simple learning algorithms
- Likely to work well for unseen test data (generalization)
- Can be converted to non-linear ones (later)

Perceptron Learning

- Multiple solutions exist for linearly separable data



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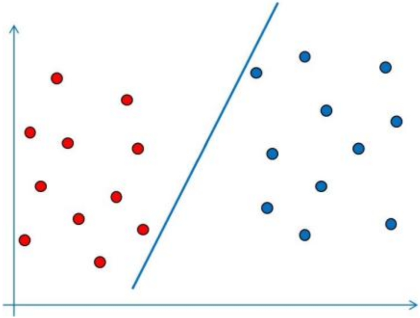
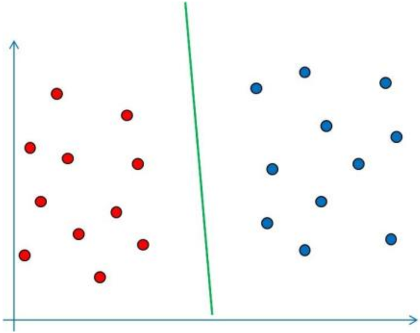


Perceptron Learning

- Multiple solutions exist for linearly separable data
- Perceptron learning (any GD) results in a feasible solution



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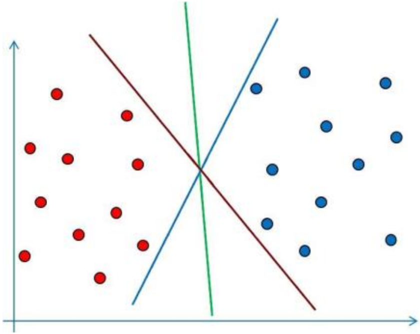
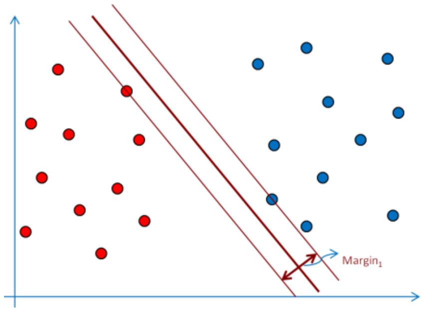


Margin: The No-mans Band

- Margin: Width of a band around decision boundary without any training samples



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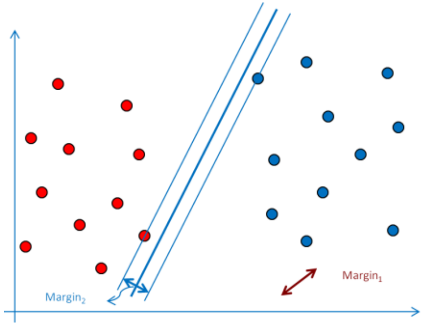

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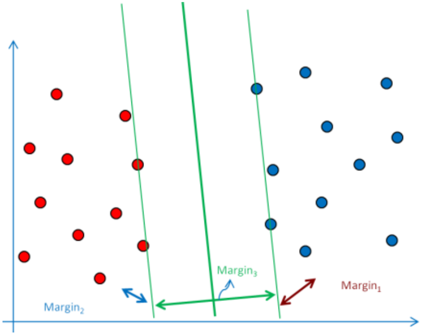
- Margin varies with the position and orientation of the separating hyperplane



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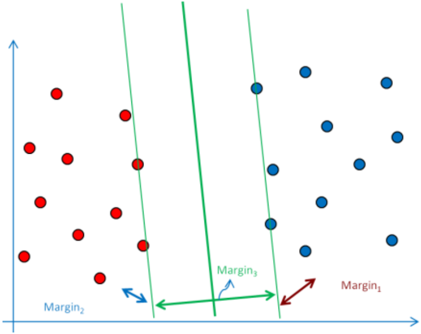
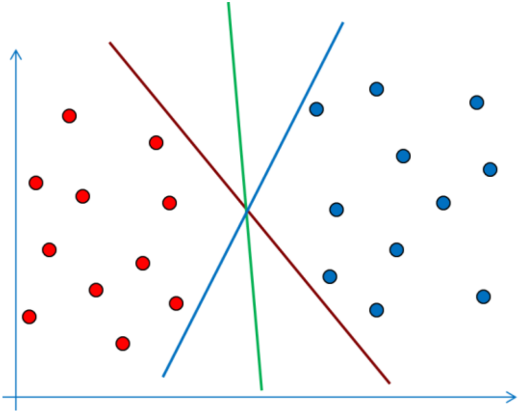


Is a Larger Margin better? Why?

Why Maximize Margin?

- Test samples vary from training data
- What is their chance of being misclassified?



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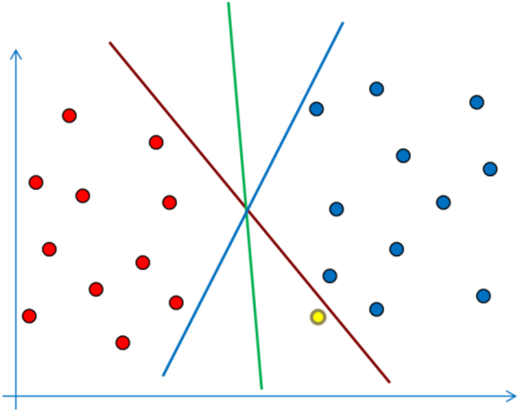

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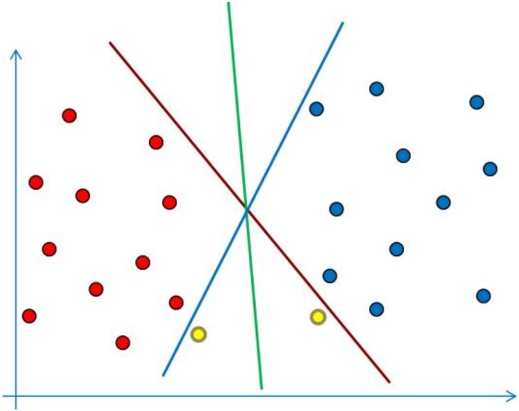

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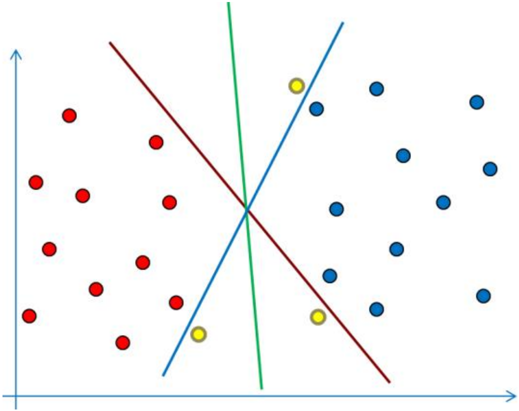

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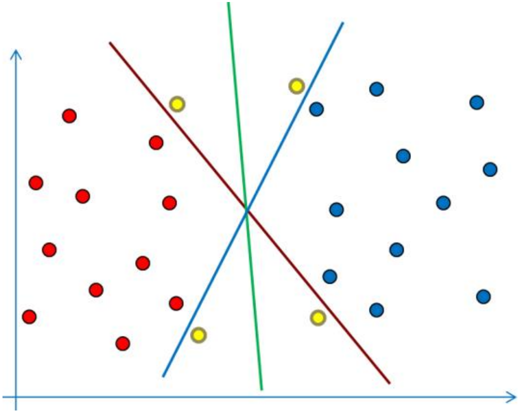

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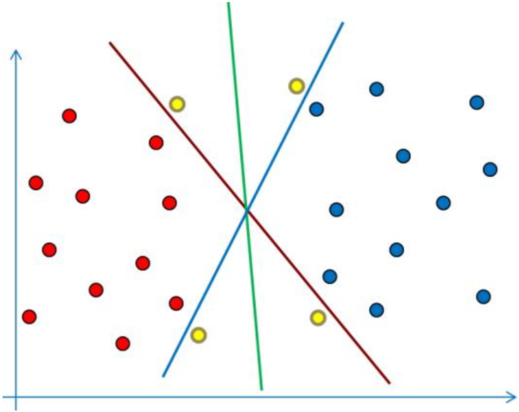
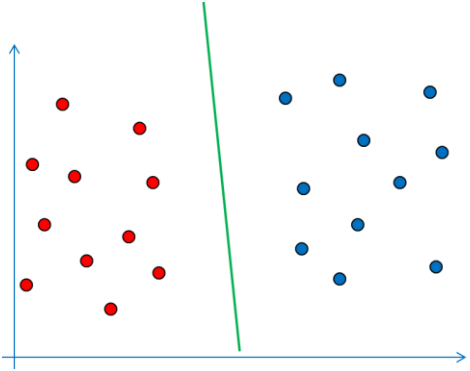
Training and Test Samples come from the same population



Summary: Max-Margin Classification

- A Large Margin will reduce the chance of misclassifying future test samples
- In other words, large-margin classifiers will generalize better.



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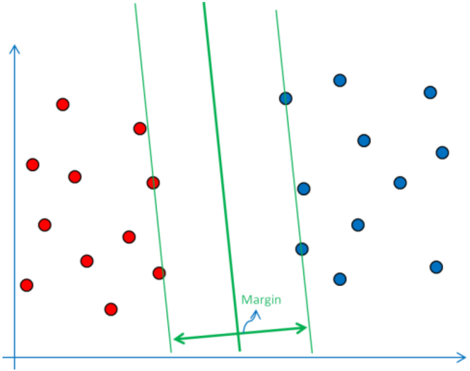

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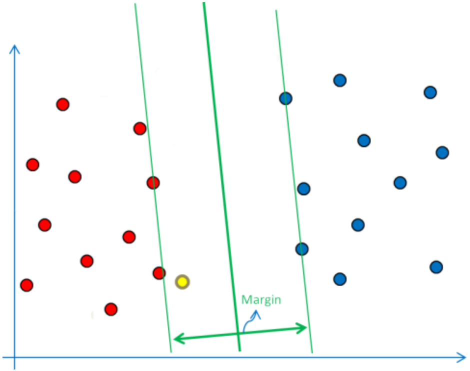

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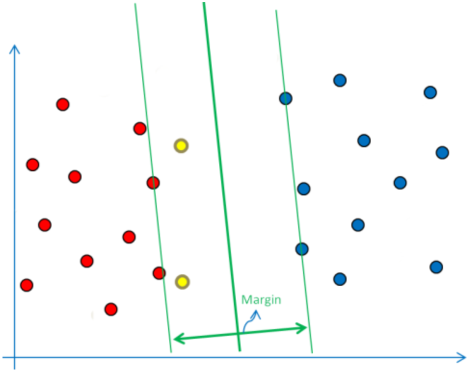

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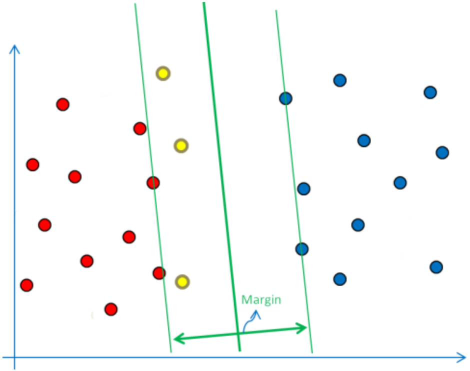

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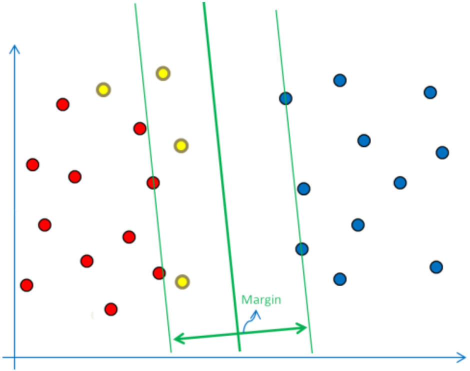

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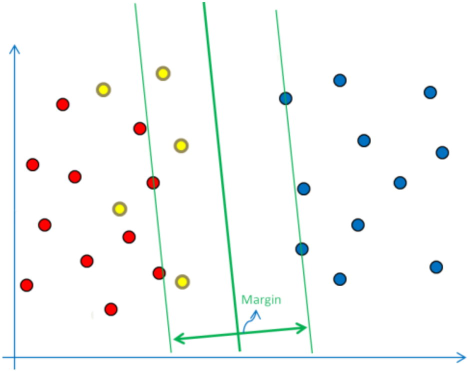

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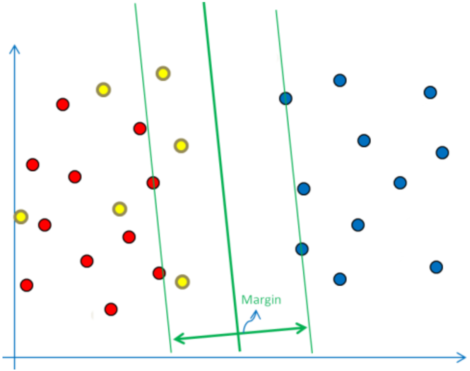

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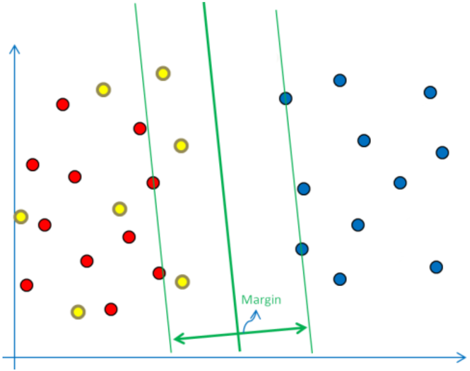

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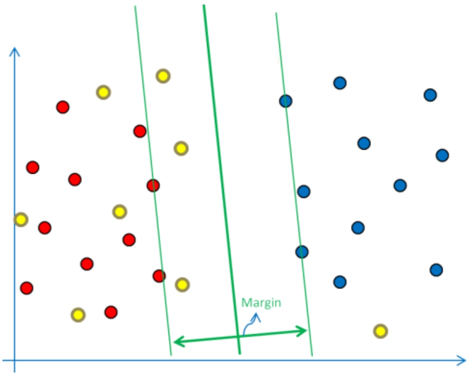

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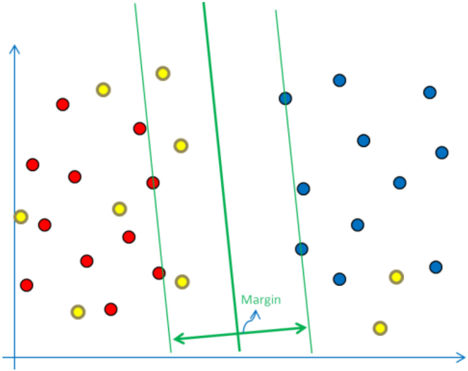

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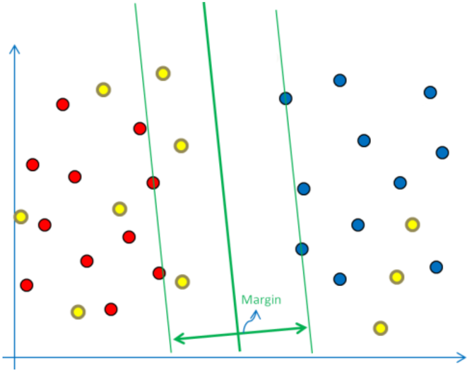

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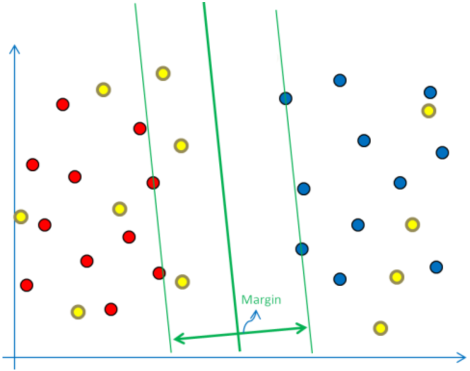

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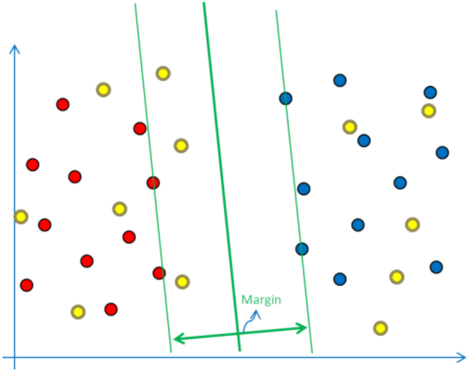

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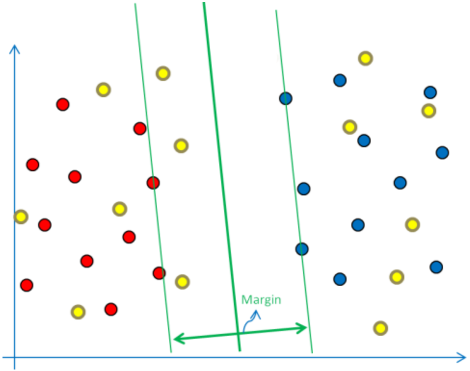

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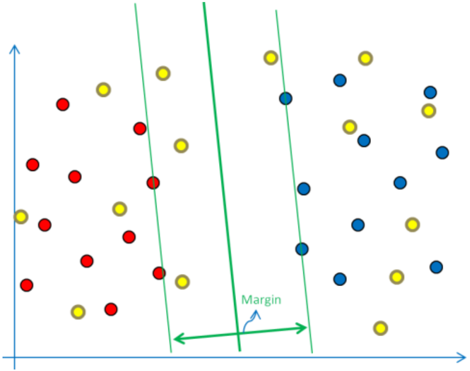

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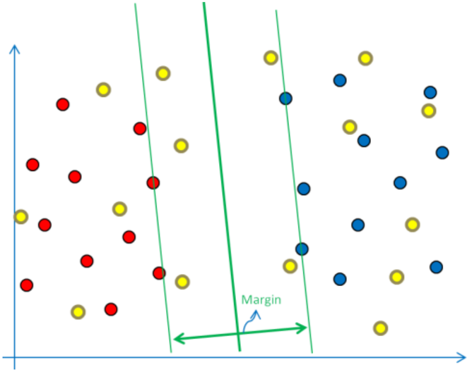

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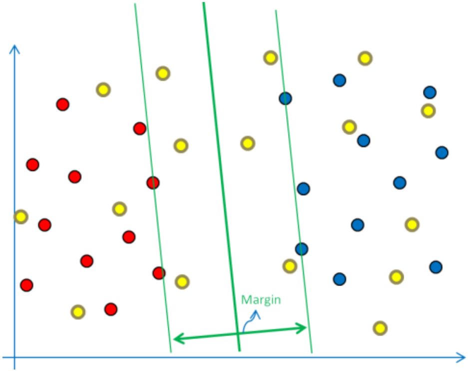

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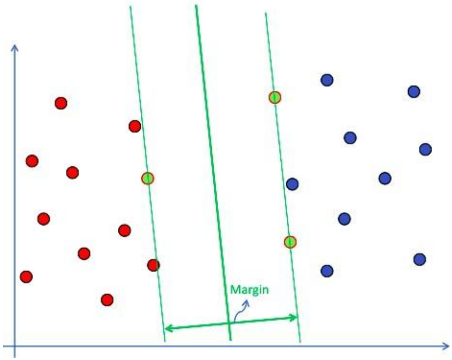

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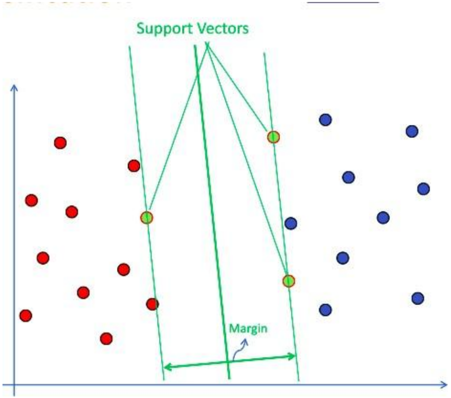

Summary

- SVMs are very good at generalization
- Convex optimization. No worries about local minima.
- Many excellent solvers. (Often we never code ourselves.)
- Linear SVMs are efficient for training and testing (both memory and flops).
- They are also highly accurate